

Chapter 131

Primaries Palette

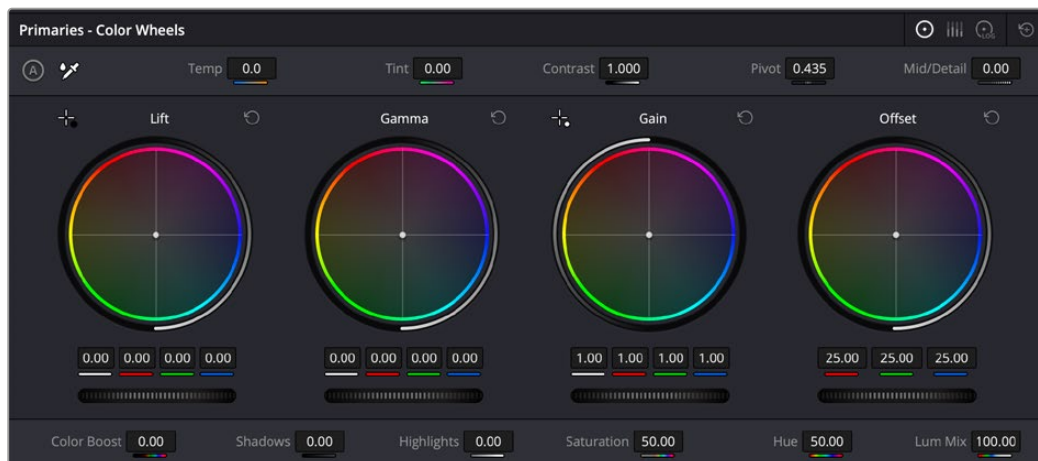
This chapter focuses on the core color adjustments that you'll be making to create “primary” corrections that alter the overall color and contrast of the image using both the Lift/Gamma/Gain adjustments in the Wheels and Bars modes of the Primaries palette, and the Shadow/Midtone/Highlight/Offset controls in the Log mode, both of which have traditionally formed the foundation of most grades.

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Introduction to the Primaries Palette

If you've had any exposure to color correction tools in any application, the Lift/Gamma/Gain controls found within the Wheels mode of the Primaries palette should look familiar. These controls correspond to the most traditional and basic color correction functionality available in DaVinci Resolve and are designed to let users without control panels have easy access to color balance and YRGB contrast manipulation using a mouse, tablet, or trackpad.



Color Balance controls for primary grading, in Wheels mode

As the name implies, the Primaries palette has been designed for the kind of overall “primary” color adjustments to the image that typically serve as the foundation of any grade. “Secondary” adjustments then refer to more specific adjustments made to isolated subjects within the image. However, it’s a time-tested practice that you do everything you can to get the overall image right using one or more primary corrections before moving on to more specific detail work built on top of these initial adjustments. For one thing, it’s a more structured, organized way to work. However, it’s also the most efficient way to make sure you’re tackling the image in a holistic way, that maximizes improvements while minimizing stress on the video signal.

The Primaries palette has three distinct modes of operation:

- Wheels mode contains the traditional DaVinci YRGB Lift/Gamma/Gain controls that allow tonally specific yet widely overlapping regions of adjustment. Wheels mode also provides access to the RGB Offset/Printer Points color balance and master controls.
- Bars mode gives access to the same YRGB Lift/Gamma/Gain and Offset controls as the Primaries Wheels mode, but the Bars interface lets you make vertical slider-driven adjustments to individual YRGB lift, YRGB gamma, and YRGB gain channels, as well as providing a slider and button-driven interface for the RGB Offset/Printer Points controls.
- Log mode is the home of the RGB Offset/Printer Points color balance and master controls that adjust the overall signal in a linear, film-oriented way. Log mode provides Shadow/Midtone/Highlight/Offset controls that offer more restrictive yet customizable regions of adjustment intended for making adjustments to log-encoded image data.

Which mode you use depends entirely on how you like to work, and what kinds of adjustments you need to make.

HDR Grading With the Primaries Palette

Starting in DaVinci Resolve 17, some of the palettes of the Color page with which color adjustments are made have been made “color space aware.” This means that the functionality of the palette will be mostly the same no matter what Timeline Color Space you’re using, or whether you’re grading to create SDR or HDR output. However, not every color palette has been made color space aware, and so when using various grading controls in the Color page to grade wide-latitude images for HDR output, if you find the controls aren’t working smoothly, you may find it useful to enable the HDR Mode of the node you’re working on by right-clicking that node in the Node Editor and choosing HDR Mode from the contextual menu (this is only available in Resolve Studio).

This setting forcibly adapts that node’s controls to work within an expanded HDR range. Practically speaking, this makes it easier to work with wide-latitude signals using controls that operate by letting you make adjustments at different tonal ranges such as Lift/Gamma/Gain or the Log controls.

Which Do I Start With, the Primaries or HDR Palette?

The HDR palette, introduced in DaVinci Resolve 17, has also been developed to work as a powerful method of creating primary adjustments to serve as the foundation of your grade. While the Global and Zones controls of the HDR palette operate from a different philosophy of signal adjustment, they’re designed to let you tackle the same issues, so it’s a completely valid choice to start out using the HDR palette instead of the Primaries palette. So, which to use?

Ultimately, this is going to come down to your comfort with the HDR palette, and your experience with the Primaries palette. If your muscle memory of working with the Primaries palette continues to make it the fastest way for you to dial in your adjustments, there’s no reason to stop using it now. In fact, the Offset/Printer Points controls in the HDR palette continue to be adjustments that are distinctly different from the functionality found within the HDR palette.

However, you should also give the HDR palette a try, because it has powerful primary grading capabilities that aren’t possible with the Primaries palette modes. Even if the functionality is new to you, if you give it a week you’ll find innumerable advantages to the HDR palette’s way of doing things, even though it requires some new ways of looking at the process.

The bottom line, however, is that you’ll probably continue using both palettes. If you do, keep in mind that in each Corrector node’s order of image processing operations, the HDR palette controls are processed before the Primaries palette controls, largely because the HDR palette’s more “photographic” approach to grading has been intentionally designed to meet the needs of primary color adjustment in our new world of wide-gamut HDR mastering and output, so it’s been fit into the image processing pipeline as the new foundation of all adjustments.

Shared Controls in the Primaries Palette

This section explains, in a generic way, how to use the common control interface conventions used by the Wheels, Bars, and Log modes to make adjustments. It also describes the use of parameters that are shared among all three modes.

Switching Between Primaries Tools

The icons in the upper right of the Primaries palette allow you to switch between the Primaries tools at the click of a button.

	Primaries: Color Wheels
	Primaries: Color Bars
	Primaries: Log Wheels

Color Balance Controls

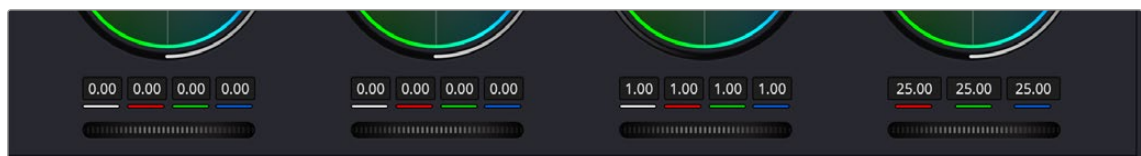
Whether you're using the Wheels or Log controls, Color Balance controls provide a way to adjust all three color channels of a particular range of the image simultaneously with a single move of the pointer, according to the mode that's currently selected. There are also a variety of keyboard modifiers that let you make specific adjustments via the GUI. These controls also correspond to the trackballs found on any of the DaVinci or third-party control panels, should you have one connected.

Methods of making adjustments using the Color Balance controls:

- **Click and drag anywhere within the color ring:** Moves the Color Balance indicator relative to its previous position, and rebalances the three color channels in whatever range of image tonality is governed by that control. You don't need to drag the Color Balance indicator itself. This simulates the kind of relative control you get when using a trackball to manipulate these parameters. As the Color Balance indicator moves, the RGB parameters underneath change independently to reflect the independent adjustments being made to each channel.
- **Shift-click and drag within the color ring:** Jumps the Color Balance indicator to the absolute position of the pointer, letting you make faster, more extreme adjustments to the color balance governed by that control.
- **Double-click within the color ring:** Resets the color adjustment without resetting the corresponding contrast adjustment for that control.
- **Command-click and drag within the color ring:** Adjusts YRGB contrast identically as if you were dragging that control's master ring.
- **Click the reset control at the upper-right of a color ring:** Resets both the Color Balance control and its corresponding master ring.

Master Wheels

The Master Wheels, located below the Color Balance controls, let you adjust the YRGB channels for a particular range of image tonality together. This has the practical effect of letting you adjust image lightness and contrast.



Lift, Gamma, and Gain or Shadow, Midtone, and Highlight, plus Offset master wheels to adjust contrast

The Master Wheels correspond to the rings surrounding the trackballs on all of the DaVinci control panels, which let you modify image contrast via YRGB adjustment (as opposed to modifying image contrast via Y-only adjustment, discussed later in this chapter).

To adjust a Master Wheel:

Dragging a Master Wheel to the left makes the corresponding tonal region of the image darker, and dragging it to the right makes that tonal region of the image lighter. The effect will vary according to the mode you're in. As you make an adjustment, the YRGB parameters located underneath all change together to reflect the simultaneous adjustment you're making to all channels.

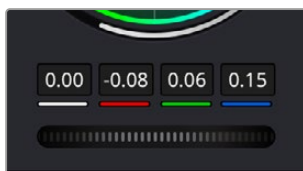
Numeric Parameters

Each Color Balance Control and Master Wheel pair of controls also have a set of four YRGB number fields underneath that display the YRGB adjustment being made by both controls. These four values encompass every color and master adjustment you can make with these controls, and they also directly correspond to the Bars interface that mirrors the Wheels controls.

These fields can be edited like any other parameter on the Color page. Even though these values display two decimal places of precision because of space restrictions in the interface, they really contain three decimal places of precision since these are floating point operations; you just can't see the third decimal place.

To edit YRGB values directly:

- You can double-click on a field to edit its value numerically.
- You can insert the text cursor next to a value in this field, and use the Up and Down Arrow keys to adjust the value one digit at a time.
- You can cut, copy, and paste values among fields.
- You can click on the field and drag left or right to adjust its value with a virtual slider.



The number fields for each Color Balance Control and Master Wheel are editable.

Shared Adjustment Controls

The three modes of the Primaries palette also share two strips of controls for making more specific adjustments to different aspects of the image, such as Contrast, Saturation, Hue, Highlight retrieval, Color boost, and so on.

Like most parameters in DaVinci Resolve, clicking and dragging a parameter's name or value to the left or right lowers and raises that parameter with a virtual slider, while double-clicking that parameter's number lets you edit it numerically, and double-clicking that parameter's name resets the parameter to its default position.



The top adjustment controls

- **Temp:** A specifically constrained Gain color balance adjustment that lets you adjust the image along a warm/orange to cool/blue axis corresponding to the naturalistic spectrum of color temperatures used for lighting. Raising this parameter performs a Gain color balance adjustment toward orange, while lowering this parameter to a negative value performs a Gain color balance adjustment toward a blue/cyan split. 0 is unity. The range is -4000 to $+4000$.
- **Tint:** A specifically constrained Gain color balance adjustment that lets you adjust the image along a magenta to green axis corresponding to the unnatural spectrum of color temperatures found in artificial lighting sources such as fluorescent and sodium vapor lighting fixtures. Raising this parameter performs a Gain color balance adjustment toward magenta (sometimes referred to as “minus green” to correct for fluorescent lighting), while lowering this parameter to a negative value performs a Gain color balance adjustment toward green (“plus green” to correct for other kinds of lighting). 0 is unity. The range is -100 to $+100$.
- **Contrast:** The Contrast parameters let you quickly narrow or widen image contrast about a user-definable pivot point. Regardless of which mode you’re in, these parameters are identical. This one parameter lets you increase or reduce the distance between the darkest and lightest values of an image, raising or lowering image contrast. The effect is similar to using the Lift and Gain master controls to make simultaneous opposing adjustments. Bright and dark parts of the image are pushed apart or brought together about a center point defined by the Pivot parameter. The “Use S-curve for contrast” setting in the General Options panel of the Project Settings (on by default) sets the contrast control to apply an “S-curve” to the image, such that the shadows and highlights of a signal will not be clipped when you increase the value. If you would prefer for these contrast adjustments to be made linearly, and for the signal to be allowed to clip when you reach the upper and lower boundaries of the video signal, you can turn this checkbox off.
- **Pivot:** Changes the center of tonality about which dark and bright parts of the image are stretched or narrowed during a contrast adjustment. Darker images may require a lower Pivot value to avoid crushing the shadows too much when stretching image contrast, while lighter images may benefit from a higher Pivot value to increase shadow density adequately.
- **Midtone Detail (MD):** When this parameter is raised, the contrast of regions of the image with high edge detail is raised to increase the perception of image sharpness, sometimes referred to as definition. When this parameter is lowered to a negative value, regions of the image with low amounts of detail are softened while areas of high-detail are left alone. 0 is unity. The range is -100 through $+100$.



The bottom adjustment controls

- **Color Boost:** A non-uniform saturation operation that affects regions of low saturation more than regions of high saturation. This is sometimes referred to as a vibrance operation. 0 is unity, showing the original color values. Raising color boost from 0-100 increases color intensity, but low-saturation parts of the image are raised more aggressively. Lowering color boost from 0 to -100 decreases color intensity, but low-saturation parts of the image are lowered more aggressively. 0 is unity, showing unaltered saturation. The range is -100 through $+100$.
- **Shadows:** Lets you selectively lighten or darken shadow detail. Raising this value retrieves shadow detail recorded below 0 percent, while leaving the midtones alone. 0 is unity. The range is -100 through $+100$.

- **Highlights:** Makes it easy to selectively retrieve blown-out highlight detail in high-dynamic-range media by lowering this parameter, and achieves a smooth blend between the retrieved highlights and the unadjusted midtones for a naturalistic result. 0 is unity. The range is –100 through +100.
- **Saturation:** A uniform saturation operation that raises (above 50) or lowers (below 50) the color intensity of every color value within the image. 50 is unity, showing unaltered saturation. The range is 0 (completely desaturated) through +100 (saturation is doubled).
- **Hue:** Rotates all hues of the image around the full perimeter of the color wheel. The default setting of 50 shows the original distribution of hues. Raising or lowering this value rotates all hues forward or backward along the hue distribution as seen on a color wheel.
- **Lum Mix:** Lets you control the balance between YRGB contrast adjustments you've made using the Master Wheels or ganged Custom curves, and Y-only adjustments to contrast made using the Y channel Lift/Gamma/Gain controls of the Primaries palette or the unganged Luma curve. At the default of 100, YRGB and Y-only adjustments to contrast contribute equally. Reducing this value diminishes the effect of Y-only contrast adjustments until, at 0, Y-only contrast adjustments are turned off.

Additionally, you'll notice that at the default Lum Mix setting of 100, individual adjustments to R, G, or B using the RGB sliders or unganged Custom curves result in automatic adjustments being made to the other two color channels in order to maintain constant Luma levels. At a Lum Mix setting of 0, individual color channel adjustments have no effect on the other color channels.

Auto Correction

The Auto Color command provides a quick way to automatically balance the blacks and whites of a clip based on the current frame at the position of the playhead. As of DaVinci Resolve 16, the A button in the Primaries palette and the Shot Match command that is available from the Thumbnail Timeline contextual menu both now use advanced algorithms, based on the DaVinci Neural Engine, to provide superior results when automatically adjusting color balance and contrast. These controls have been developed to provide optimal results when working in the Rec. 709 color space, and at a gamma of 2.4, so they work well in conjunction with using Resolve Color Management (RCM) to normalize media first.

For more information on using these, see *Chapter 127, "Automated Grading Commands and Imported Grades."*

Reset Controls

The Reset control, found at the upper right-hand corner of the Primaries palette, lets you reset every single setting in the entire palette. However, there are a variety of parameter-specific reset controls available to make more targeted resets.

- Each pair of Color Balance and Master Wheel controls have a reset control that resets both.
- Each numeric parameter can be individually reset by double-clicking the name of the parameter.
- The numeric parameters underneath each Color Balance and Master Wheel control pair can be reset by double-clicking the color label that appears underneath them.

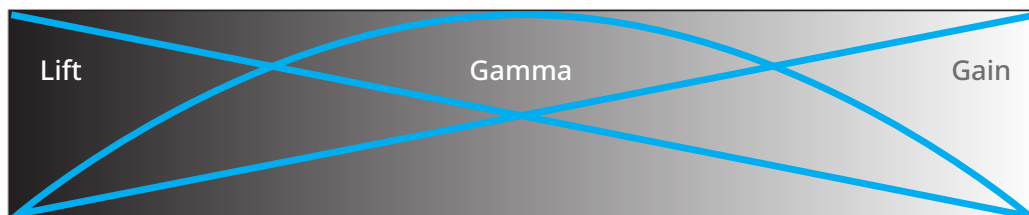
Color Wheels and Bars

The Color Wheel palette's Primaries Wheels mode lets you rebalance color and adjust contrast via the traditional DaVinci controls, which govern three overlapping tonal ranges referred to as Lift, Gamma, and Gain. The Lift/Gamma/Gain color balance and Master Wheel controls are tied to the YRGB Lift/Gamma/Gain sliders found in the Primaries palette; adjustments made to one set of controls are mirrored in the other.



Though they may look different, the Wheels and Bars controls both actually adjust the same components, but in different ways.

These tonal ranges are defined by image lightness, on a scale where 0 is absolute black and 1023 is absolute white. The following illustration shows an approximation of how the Lift, Gamma, and Gain tonal zones broadly overlap, and how each zone's influence falls off towards the opposing extremes of image tonality.



Graphic displays the relationship of the Lift, Gamma, and Gain controls over the image brightness range that they control

The Lift color balance control region of influence starts at black, and then falls off through the middle grays to diminish to no influence at white. Meanwhile, the Gamma color balance control has its greatest influence over the image in the middle grays, and its influence diminishes towards black and white. Lastly, the Gain control is the inverse of Lift, having its greatest effect on the image at white, with its influence falling off to diminish at black.

Because these tonal ranges overlap so broadly, you can make very soft, subtle, naturalistic adjustments using these controls. Furthermore, you can capitalize on their overlap by moving an adjacent color balance control toward a color that's complementary to an adjustment you've just made to restrict further how much of the image is affected.

The following image shows the interaction of extreme corrections made to a grayscale image using all three Color Balance controls. Lift has been pushed toward green, Gamma has been pushed toward blue, and Gain has been pushed toward red.



Extreme adjustments showing the overlap of the Lift, Gamma, and Gain color balance controls

Notice how, even though these corrections are extreme, the colors blend smoothly. This is the reason for the broad overlap among all three controls, and why Lift, Gamma, and Gain are so effective in making corrections to the ambient color temperature of a scene to account for inconsistencies in lighting or camera settings.

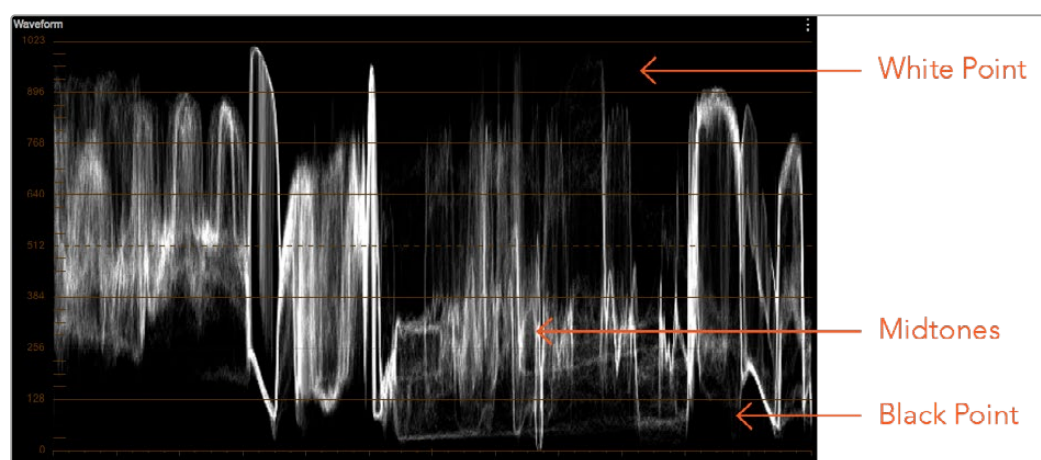
3-Way Master Wheel Adjustments

The Master Wheels located below the Color Balance controls let you precisely modify image contrast by YRGB adjustments, which individually alter the black point, the white point, and distribution of midtones that fall in between.

Lift: Lets you adjust the perceived shadow density of the image by altering the black point of the image. Dragging the Lift master wheel to the left makes the darkest values in the image darker, increasing the distance between the black and white points of the image, and stretching all the midtone values in-between. Dragging the Lift master wheel to the right makes the darkest values in the image lighter, reducing contrast and squeezing all the midtone values between the black and white points.

Gamma: Lets you adjust the overall perceived lightness of the image by altering the distribution of midtones that fall between the Lift and Gain master wheel settings. Dragging the Gamma master wheel to the left darkens the overall image, while dragging it to the right brightens it. Most Gamma contrast adjustments have a minimal effect on the black and white point of the image, but large adjustments may push either boundary of image lightness farther out. This interaction is described in more detail below.

Gain: Lets you adjust the lightness of the highlights by altering the white point of the image. Dragging the Gain master wheel to the left makes the lightest values of the image darker, squeezing the midtones between the white and black points of the image. Dragging Gain to the right makes the lightest values even lighter, eventually clipping at maximum white.



Waveform display shows the clip's contrast range

These contrast adjustments are not limited by one another. For example, raising or lowering the Gamma master wheel by a large amount may push the highlights of the image higher or the shadows of the image lower, regardless of the current Lift or Gain contrast setting.

As a result, these controls are somewhat interactive, and you may find yourself going back and forth between controls as you make your final contrast adjustments. This is one of the reasons a control panel is valuable, as it allows you to adjust all three settings simultaneously.

Offset Color and Master Controls

The fourth set of Color Balance and Master Wheel controls is actually shared with the Log controls and with the Offset sliders in the Primaries palette. These are the Offset controls, which let you make linear adjustments to rebalance the entire tonal range of the RGB channels. There is no Y-only control for Offset, only a Master RGB adjustment.

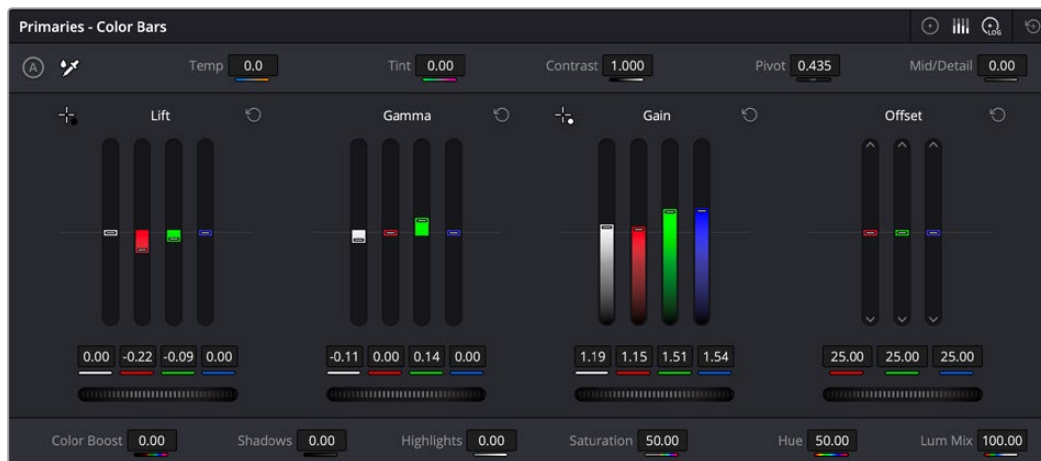
The Offset color balance control: Works as a simultaneous adjustment to all three Offset sliders located in the Primaries palette; adjustments made to the Offset color balance control also alter the Offset sliders. Used subtly, this makes it easy to neutralize color imbalances in the darkest part of the image, while simultaneously rebalancing every other part of the image. Used more dramatically, this control makes it easy to add a color wash throughout the entire image.

The Offset master wheel: Acts as a global adjustment to image lightness, an operation sometimes referred to as setup, raising or lowering all RGB channels together.

When using a DaVinci Micro or Mini control panel, the Offset color balance and master controls are adjusted via either the third trackball and ring while in Offset mode, or the fourth trackball and ring of the DaVinci Advanced Panel in Offset mode.

Color Bars Mode

The Bars mode contains the original set of DaVinci Resolve color adjustment sliders. These sliders serve two uses. First, they're highly visible indicators of the individual YRGB channel adjustments that are made using the trackballs, rings, and knobs of a grading control panel. Second, they provide control of individual YRGB Lift/Gamma/Gain parameters using a mouse, tablet, or trackpad.



Primary grading controls

The main controls of the Bars mode are the individual Luma (or Y), Red, Green, and Blue sliders, four each for Lift, Gamma, and Gain. These sliders provide precise Lift/Gamma/Gain style control over each of the YRGB channels of the image. When used in conjunction with a Parade Scope video analysis of the image, these controls can enable you to fix irregular color imbalances in specific channels quickly.

Additionally, the Luma (Y) Lift/Gamma/Gain sliders allow easy Y-only adjustments to contrast, where an increase in contrast results in perceptually diminished color saturation. These correspond to the three knobs arranged vertically to the left of the Trackball panel of the DaVinci Advanced control panel, or the three left-most knobs of the DaVinci Micro or Mini control panels.

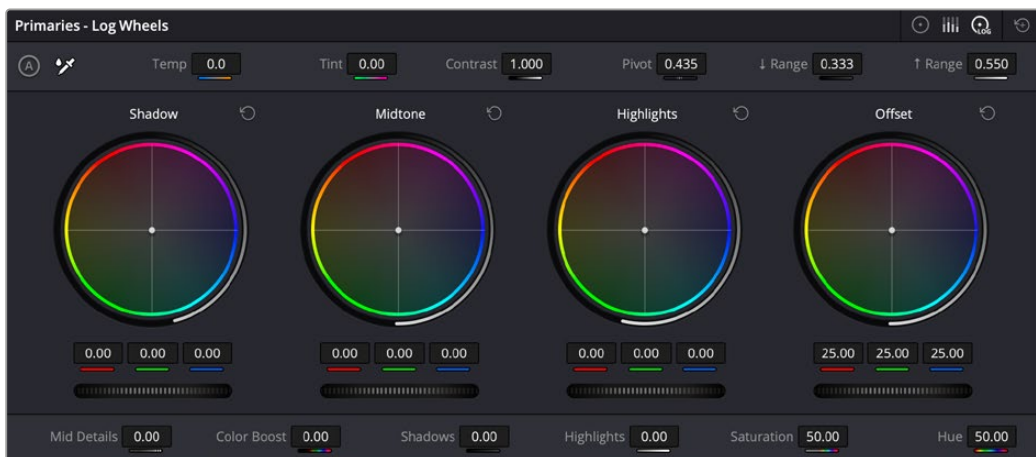
Making Y-only adjustments to contrast is a great way to increase contrast when you're going for muted saturation or a gritty look. This kind of adjustment is also useful in situations where you're trying to increase shadow density without increasing image colorfulness.

Log Wheels Mode

The Shadow/Midtone/Highlight color balance and Master Wheel controls operate independently of the Lift/Gamma/Gain color balance and Master Wheel controls found in the Primaries mode. While the Log mode uses the same types of controls as the Primaries mode, the way each control affects the image is very different.

To switch between the Primaries and Log modes of the Wheels:

Choose an option from the mode drop-down, click the right mode button, or press Option-Z.



Log color balance controls, with behavior that is very different from the Wheels mode color balance controls

There are two ways of using the Log mode controls. The first takes advantage of the way these controls work to make fast, filmic adjustments to log-encoded media prior to it being normalized or “de-logged” by operations that are performed using nodes appearing after it in the image processing pipeline of your node tree. Normalizing or de-logging the image can be done via Color Space Transform operations, LUTs, and manual adjustments if you’re grading using DaVinci YRGB color science. If you’re using color management, they can be done via the Output Color Space setting of Resolve Color Management (RCM), or via the ACES Output Device Transform (ODT).

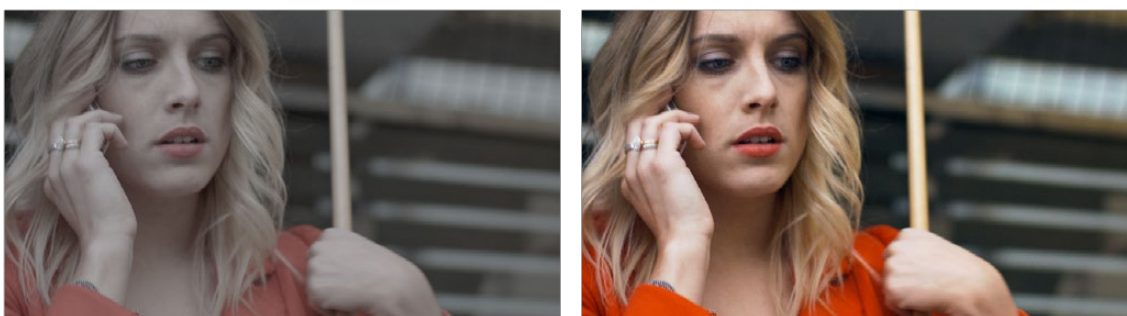
The other way of using the Log controls is to take advantage of the more restrictive, but adjustable tonal range of the Shadow/Midtone/Highlight controls to stylize normalized clips by tinting or adjusting the contrast of tonally-specific regions of the image.

Whether or Not to Use Legacy Log Grading Ranges and Curve

DaVinci Resolve 12.5 introduced a modification to the Log grading controls that provides smoother, more pleasing results using the same controls. To maintain backward compatibility with older projects, a “Use legacy Log grading ranges and curve” checkbox in the General Options panel of the Project Settings lets you switch your project between the older Log control behavior and the newer one. Older projects that are opened in DaVinci Resolve have this checkbox turned on by default, while new projects have this turned off by default.

Using the Log Mode Controls to Grade Log-Encoded Media

The Log controls are so named because they’re designed to work specifically with media with Log-C or similar gamma and color encoding, derived from the Cineon Log gamma curve, developed by Kodak to digitally store flat-contrast, wide-gamut image data that preserves image detail with a wide latitude for adjustment.



An example of a log-encoded clip (left), and the same clip after being normalized (right)

You can debayer most raw formats to a log-encoded image in order to derive the maximum amount of image data and adjustable latitude from that source.

For more information, see *Chapter 7, “Camera Raw Settings.”* However, the resulting image needs to be normalized to occupy the final range of color and contrast that you intend for the final result.

You can do this one of two ways:

- You can make a very careful curves adjustment in a second node to stretch the log-encoded image out to fit the contrast profile you want. By making this adjustment in Node 2, you make room for a customizing adjustment made using the Log controls in Node 1, prior to the normalization adjustment. This is key.
- You can also apply a 1D Output or 3D LUT to the first node of a clip to normalize the image. This is a faster, if less flexible operation, but a smooth tonal range may be easier to obtain. Since a LUT applied within a node is always the last adjustment within that node’s order of operations, you can also use Node 1’s Log controls to customize the look of the footage.

In either case, it’s important that the normalizing adjustment happens after your Log control adjustments, for the Log control adjustments to work as they should. With your node tree set up in this way, you’ll be monitoring an ordinary-looking image, but taking advantage of the Log mode controls’ unique tonal ranges to manipulate the log-encoded image data with great specificity.

TIP: Within corrector nodes, LUTs are applied after Log control adjustments, so if you want to keep the number of nodes in your node tree to a minimum, you can apply a normalizing (or de-logging) LUT to the same node with which you’re making Log adjustments, knowing that the LUT will be properly applied after log adjustments.

For more information, see *Chapter 144, “Image Processing Order of Operations.”*

When using the Log mode controls, here’s a workflow to consider as you learn how they work:

- First, use the Offset master wheel to set the black point, and use the Contrast and Pivot parameters to stretch or compress contrast as necessary to achieve the tonal range you require.
- Second, use the Offset color balance control to adjust the overall color balance of the image to your liking.
- Third, use the Shadow/Midtone/Highlight color balance and Master Wheel controls to make specific, targeted adjustments to the color and contrast of the image in tonal ranges that match where that data is in the log-encoded image.

Working in this way, you’ll find that adjustments made with the Offset color balance and Master Wheel controls and Contrast controls control the log-encoded image very nicely to create an overall adjustment, while the Shadow, Midtone, and Highlight controls allow you to fix specific issues, such as shadow balance and density, after your main adjustment has been set.

The following illustration shows an approximation of how the default ranges of the Shadow, Midtone, and Highlight controls divide the tonal range of a log-encoded image.



This graphic shows the tonal range of each of the Log controls when used with a log-encoded image

As you can see, when used with a log-encoded image the color interactions between each adjustment overlap very softly, while still allowing more specific adjustments than those made by the Lift/Gamma/Gain controls.

Furthermore, the boundaries of the Shadows, Midtones, and Highlights Log controls can be customized using the Low and High Range parameters. This gives you added flexibility to apply more specific contrast and color adjustments.

Once you've made an adjustment using Log mode controls along with a normalizing LUT or curve adjustment, you can always apply additional nodes and use the Wheels mode of the Primaries palette to make further alterations to the now normalized image, working as you normally would with any of the other tools in DaVinci Resolve.

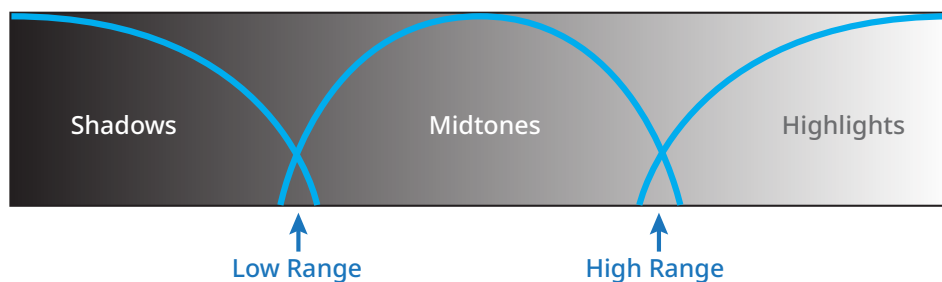
TIP: Internal to each node's image processing order of operations, the Lift/Gamma/Gain adjustments of the Wheels mode are applied prior to the Shadow/Midtone/Highlight/Offset adjustments of the Log mode, so if you want to apply Log mode controls before Wheels mode controls, then you need to use Wheels controls in a subsequently added node.

For more information, see *Chapter 144, "Image Processing Order of Operations."*

Using the Log Mode Controls to Stylize Normalized Media

You can also use the Log mode controls on already normalized images. Although the results you get will be somewhat different, this can be a fast way to create interesting stylizations. Whereas the broadly overlapping tonal ranges of the Lift, Gamma, and Gain color balance controls allow subtle adjustments to be made very easily, the Log palette's color balance controls affect much more restrictive tonal zones that overlap much less when used on normalized images.

The following illustration shows an approximation of how the Shadows, Midtones, and Highlights, by default, divide the tonal range of the image into non-overlapping thirds. As you saw in the previous section, these divisions were originally intended to map to log-encoded media. However, with normalized media these divisions provide a different, and potentially useful, set of ranges from the Primaries mode controls.



This graphic shows an approximation of the tonal range of each of the Log controls with the additional low and high range to expand or contract the range.

With normalized media, the Shadows really do only affect the darkest shadows, falling off at approximately the bottom third of image tonality. The Midtones affect only the middle third of grays, and the Highlights affect the brightest top third of image values. However, these default ranges of image tonality can be adjusted using the Low Range and High Range controls, which are described in more detail later.

The following image shows the default interaction of extreme corrections made to a grayscale image using the Log mode controls. The Shadows have been pushed toward green, the Midtones have been pushed toward blue, and the Highlights have been pushed toward red.



Extreme adjustments showing the overlap of the Shadow, Midtone, and Highlight color balance controls

As you can see, with normalized media the color interaction between each adjustment is very subtle. The darkest shadows end up bright green, the midtone values are vivid blue, and the highlights are almost pure red. This restrictiveness is useful when you want to limit a correction to a specific tonal range within the image without needing to use a Luma qualifier. It's also quite useful for making bold, stylistic color adjustments when creating a non-naturalistic look.

TIP: The Highlights adjustment of the Log controls can also be extremely useful for boosting or otherwise controlling the highlights of HDR grades. However, the multiple highlight zones of the HDR palette afford a much higher degree of control over this kind of adjustment, using similar control panel-friendly controls.

Adjusting the Default Tone Ranges in Log Mode

When in Log mode, two additional parameters appear in the top shared controls bar that let you modify the range of Shadows and Highlights image tonality that each Color Balance control affects, in turn narrowing and widening the range of Midtones. Keep in mind that while the ranges can be customized, the amount of overlap between each range cannot.

Low Range: Moves the border where the Shadows and Midtones meet. Lowering this parameter widens the range affected by the Midtones, and narrows the range affected by the Shadows. Raising this parameter narrows the Midtones and widens the Shadows.

High Range: Moves the border where the Midtones and Highlights meet. Lowering the High Range parameter narrows the range affected by the Midtones, and widens the range affected by the Highlights. Raising this parameter narrows the Highlights and widens the Midtones.

There are also Saturation and Hue parameters which mimic these settings found within other palettes and modes.

Adjusting Contrast in Log Mode

When using the Log mode controls, your primary tools for adjusting image contrast will usually be the Offset master wheel and Contrast and Pivot parameters. Using these three controls, you can set a black point and adjust the overall contrast very quickly.

The Shadow, Midtone, and Highlight master wheels let you adjust image lightness within the same restrictive ranges of image tonality that are defined by the Low Range, High Range, and Pivot parameters. These adjustments should appear smooth, if somewhat narrower than the Lift/Gamma/Gain controls, when used with log-encoded media. However, when used with normalized images, severe adjustments made with one master wheel may not always make a smooth transition to the next adjacent range of image lightness.

NOTE: Because these controls are so restrictive when used with normalized images, it's easy to create solarization effects by raising the shadows to be higher than the Midtones, or lowering the Highlights to fall below the Midtones, to give two examples.

Log Offset Color and Master Controls

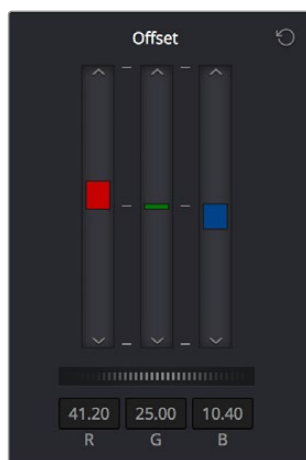
The Log controls share the same Offset color balance and Master Wheel controls that appear in the Lift/Gamma/Gain mode of the Wheels and Bars controls. The Offset controls are in fact processed as part of the Log controls, but for convenience they're presented with the Wheels controls because they're so useful.

- **The Offset color balance control:** Works as a simultaneous adjustment to all three Offset sliders located in the Primaries palette; adjustments made to the Offset color balance control also alter the Offset sliders. Used subtly, this makes it easy to neutralize color imbalances in the darkest part of the image, while simultaneously rebalancing every other part of the image. Used more dramatically, this control makes it easy to add a color wash throughout the entire image.
- **The Offset wheel:** Acts as a global adjustment to image lightness, an operation sometimes referred to as setup, raising or lowering all YRGB channels together.

Offset and Printer Points

In Bars mode, the Offset controls are represented by three vertical sliders, which mirror the settings of the Offset color balance controls in the Color Wheel palette, except that they provide individual control over the Red, Green, and Blue color channels. In fact, this interface also happens to be the epicenter of the Printer Points functionality offered by DaVinci Resolve.

When you drag one of the Offset sliders up or down, you raise or lower that color channel in its entirety. This can be useful for adjusting color channels that are particularly problematic, but it's also the way you impose the kinds of traditional linear color adjustments that color timers of film have employed for decades. In fact, many colorists consider the simplicity of these controls a virtue, and embrace the slight shadow or highlight contamination that can result from color balancing in this linear way as a creative hallmark of traditional cinema color adjustment.



Offset RGB and master controls

NOTE: Unlike the other Color Balance control adjustments described in this chapter, the Offset controls do not use the Lum Mix parameter to control whether individual adjustments to the R, G, or B channels result in automatic adjustments being made to the other two color channels in order to maintain constant Luma levels. All Offset and printer points adjustments made to a specific color channel affect only that color channel.

Adjusting Printer Points in the Bars Mode

Each Offset slider also has a pair of arrow buttons, one at the top and one at the bottom. These buttons provide “Printer Point” adjustment of these values, which let you adjust each Offset channel in discrete increments. Printer Points can be useful for projects that have tight integration with a film lab, and are designed to emulate color adjustments made using optical printers.



The “plus” and “minus” red, green, and blue buttons for making printer point adjustments

The Offset wheel control underneath the Offset sliders lets you adjust all three sliders at once, performing a master or setup adjustment. The Offset sliders, Printer Point buttons, and Offset wheel can be adjusted using keyboard shortcuts.

NOTE: The DaVinci Resolve Advanced and Mini color grading panels each have support for adjusting printer points using rotary controls for the Red, Green, and Blue channels, as well as All channels together.

Adjusting Printer Points Via Keyboard Shortcuts

One of the best ways of using printer points if you don't have a color control panel that supports them is to enable the numeric keypad to use dedicated keyboard shortcuts just for printer points.

To use the Printer Lights Hotkeys:

- Choose Color > Printer Light Hotkeys, or press Option-Command-Grave Accent (`)

When you enable Printer Light Hotkeys, there are two sets of shortcuts you can use to manipulate printer points. First, if you want to directly manipulate RGB in whole increments, there's one set for that.

Red	Green	Blue
7 - Plus Red	8 - Plus Green	9 - Plus Blue
4 - Minus Red	5 - Minus Green	6 - Minus Blue

However, if you want to work in the classic way by manipulating cyan, magenta, and yellow in whole increments, there's another set of shortcuts for that, using the remaining keys on the numeric keypad.

Cyan	Magenta	Yellow
1 - Plus Cyan	2 - Plus Magenta	3 - Plus Yellow
Minus (-) - Minus Cyan	0 - Minus Magenta	Period (.) - Minus Yellow

There are also a pair of keyboard modifiers that gives you finer control over printer points adjustments made with these special keyboard shortcuts, while retaining the coarser default adjustments that let you make bigger changes more quickly:

- Hold down Command while using these key shortcuts to adjust printer points in quarter-increments.
- You can use half-increments as well, but you will need to assign the keyboard shortcuts for these manually.

Printer Light Step Project Settings

The increments used by the Printer Point buttons are defined by the Printer Light Step Calibration parameters, found in the Color panel of the User Preferences. The default settings are designed to emulate traditional film printer adjustments, but these settings can be customized to align DaVinci Resolve's printer points adjustments with those of a particular film lab's equipment. However, if you're not working with a lab, you can change the Step and Density settings to alter how much of an adjustment each printer point makes according to your own preferences.

For more information, see *Chapter 4, "System and User Preferences."*



Controls in the Color panel of the User Preferences to adjust Printer Point functionality

TIP: If you're feeling left out because all the cool kids are using printer points and you're unfamiliar with them, making the Parade scope visible is a great way to learn how these adjustments work while seeing their specific effects on the red, green, and blue channels of the video signal.